

CMPT 732-G200 Practices for Visual Computing

Ali Mahdavi Amiri

Some history

- SVD was discovered by the following people:



E. Beltrami
(1835 – 1900)



M. Jordan
(1838 – 1922)



J. Sylvester
(1814 – 1897)



E. Schmidt
(1876-1959)



H. Weyl
(1885-1955)

Gene Golub

- SVD was significantly popularized by Gene Golub.



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MATRIX COMPUTATIONS

THIRD EDITION

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Geometric analysis of linear transformations

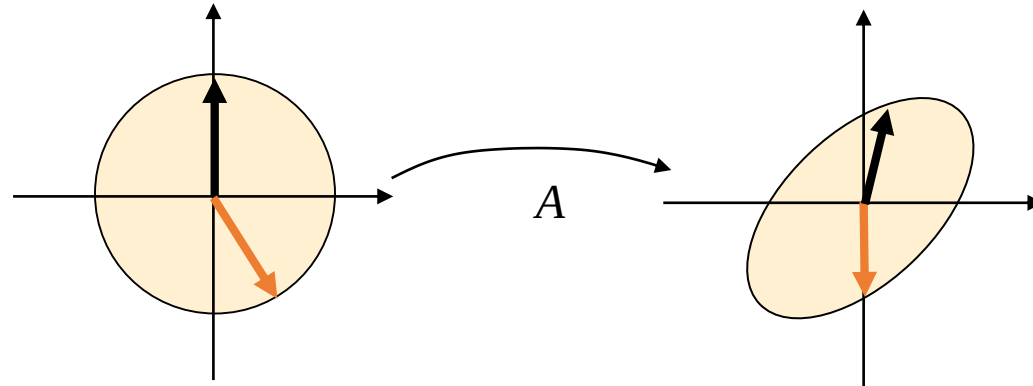
- We want to know what a linear transformation A does
 - What happens when matrix A is multiplied in the coordinates of a shape?

Geometric analysis of linear transformations

- We want to know what a linear transformation A does
- Need some simple and “comprehensible” representation of the matrix of A .

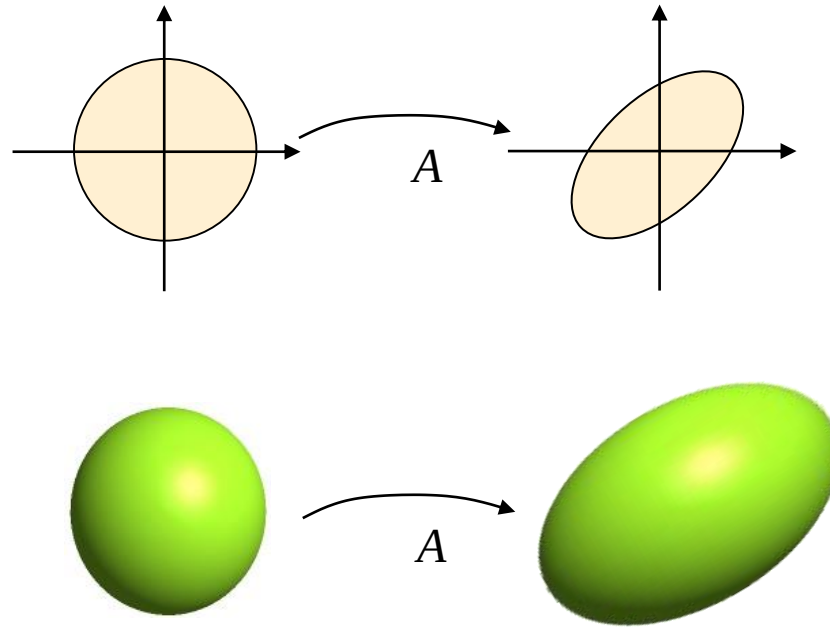
Geometric analysis of linear transformations

- We want to know what a linear transformation A does
- Need some simple and “comprehensible” representation of the matrix of A .
- Let's look what A does to some vectors
 - Since $A(\alpha \mathbf{v}) = \alpha A(\mathbf{v})$, it's enough to look at vectors $\mathbf{v}$ of **unit** length



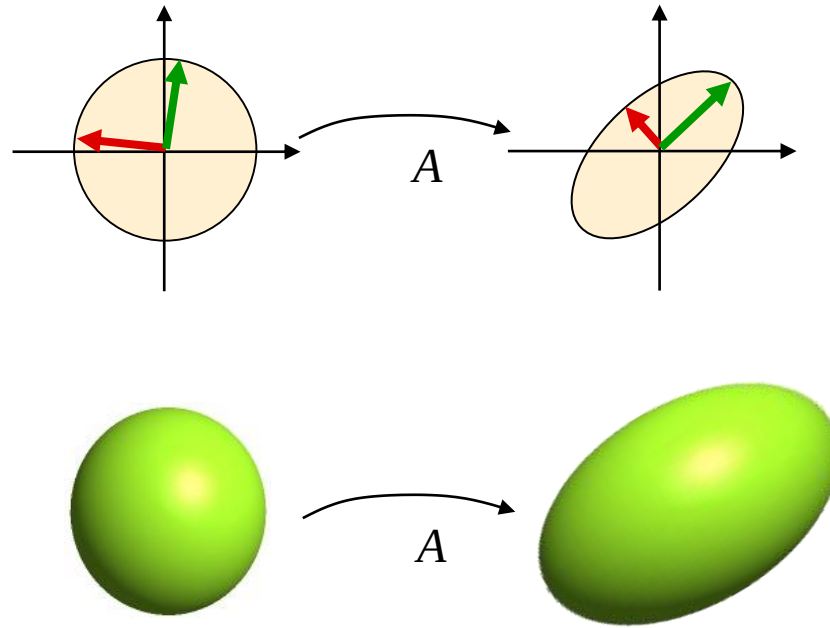
The geometry of linear transformations

- A linear (non-singular) transform A always takes hyper-spheres to hyper-ellipses.



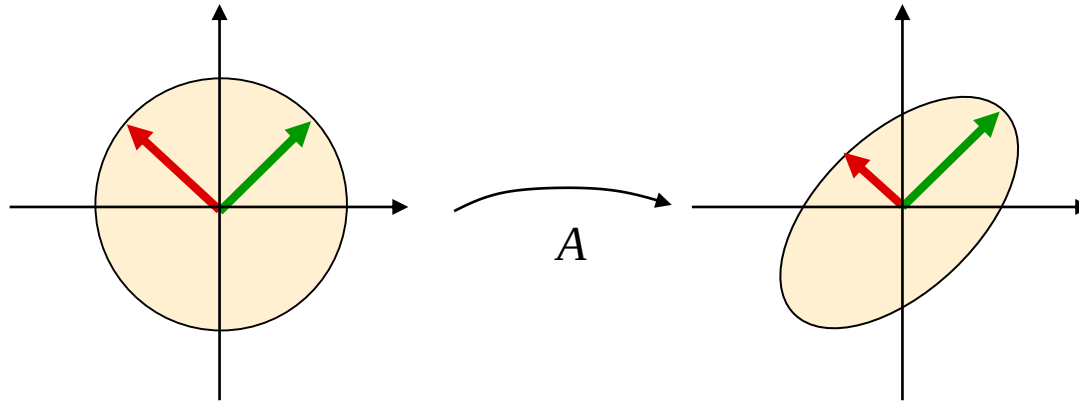
The geometry of linear transformations

- Thus, one good way to understand what A does is to find which vectors are mapped to the “main axes” of the ellipsoid.



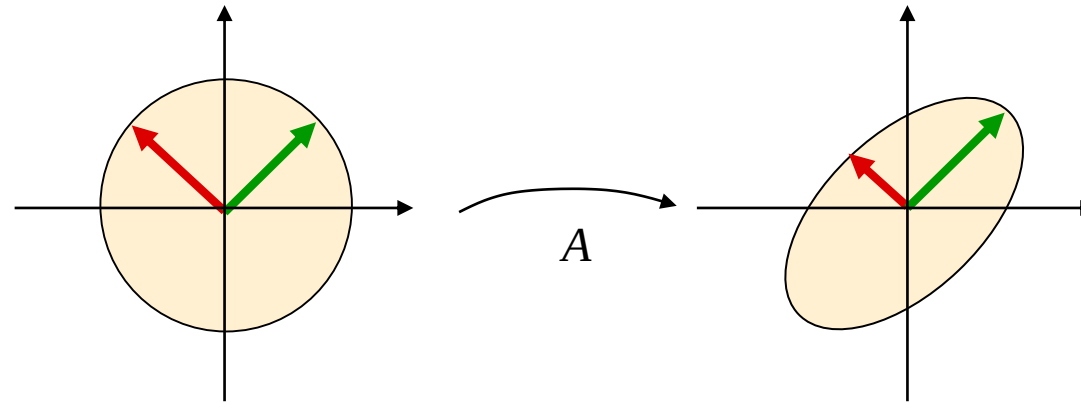
Geometric analysis of linear transformations

- If we are lucky: $A = V \Lambda V^T$, V orthogonal (true if A is symmetric)



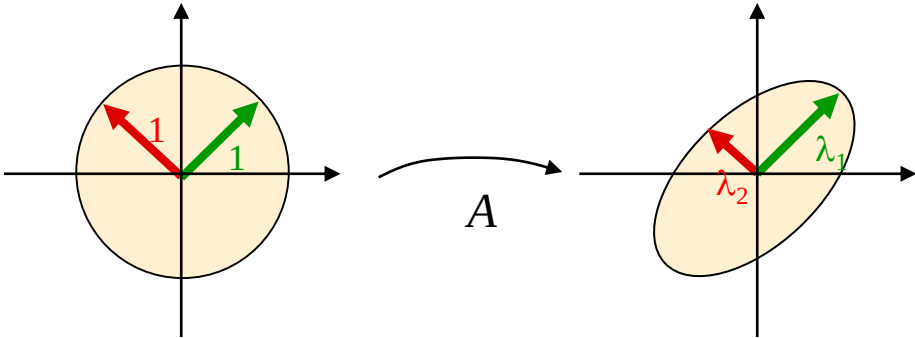
Geometric analysis of linear transformations

- If we are lucky: $A = V \Lambda V^T$, V orthogonal (true if A is symmetric)
- The eigenvectors of A are the axes of the ellipse



Symmetric matrix: eigen decomposition

- In this case A is just a scaling matrix. The **eigen decomposition** of A tells us which orthogonal axes it scales, and by how much:


$$A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n] \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]^T$$

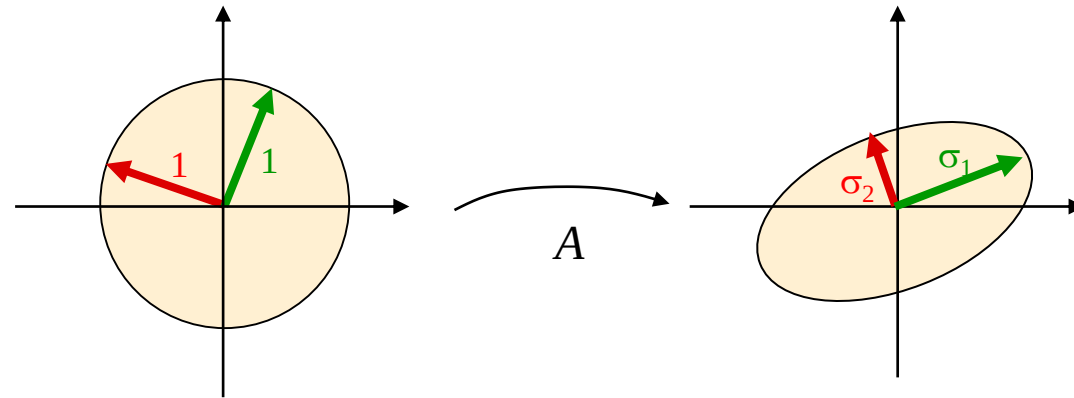
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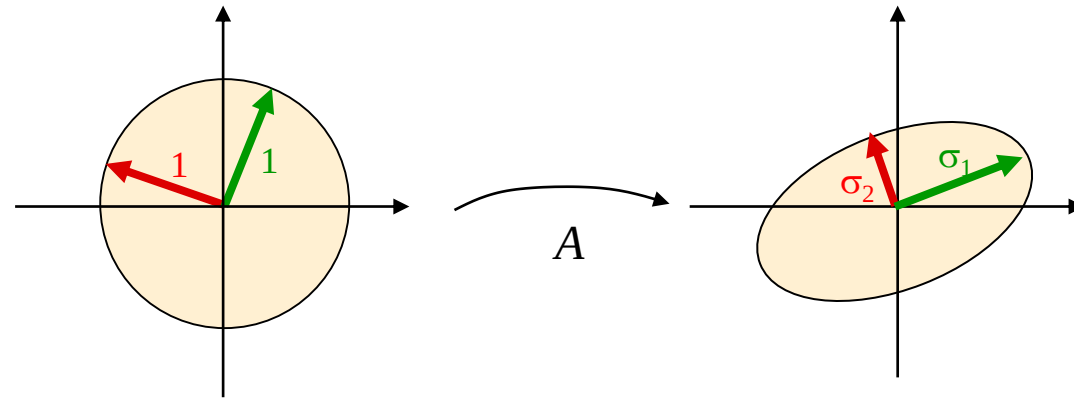
$A \mathbf{v}_i = \lambda_i \mathbf{v}_i$

General linear transformations: SVD



$$A = U\Sigma V^T \rightarrow AV = U\Sigma$$

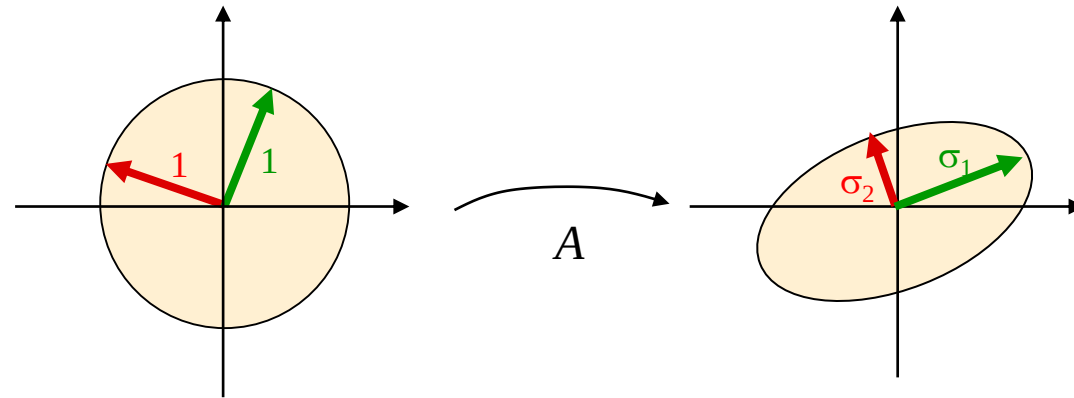
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Why?

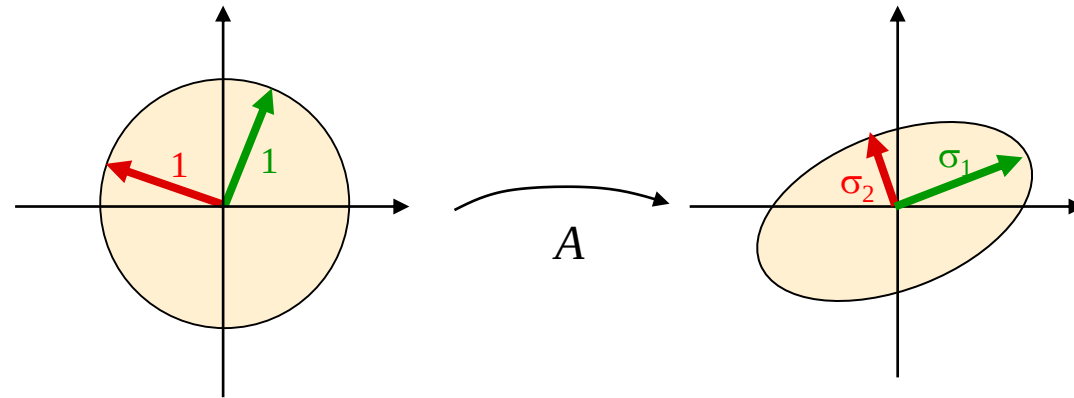
General linear transformations: SVD



$$A = U\Sigma V^T \rightarrow AV = U\Sigma$$

$$\overset{\text{orthonormal}}{A} \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \dots & \mathbf{u}_n \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix}$$

General linear transformations: SVD



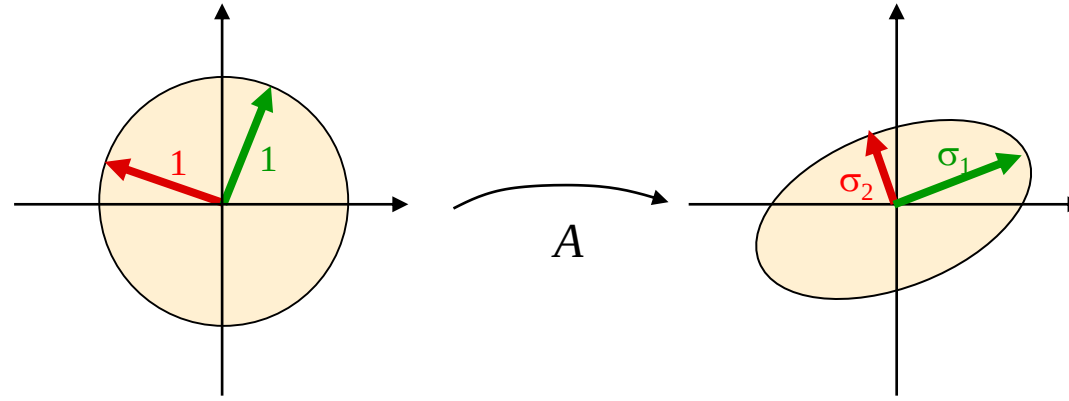
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$$A\mathbf{v}_i = \sigma_i\mathbf{u}_i, \quad \sigma_i \geq 0$$

General linear transformations: SVD

- In general A will also contain rotations, not just scales:



$$A = U \Sigma V^T$$

$$A = [\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_n] \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix} [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]^T$$

SVD is the “working horse” of linear algebra

- There are numerical algorithms to compute SVD. Once you have it, you have many things:
 - Matrix inverse $\rightarrow$ can solve square linear systems
 - Numerical rank of a matrix
 - Can solve least-squares systems
 - PCA
 - Many more...

Matrix inverse and solving linear systems

- Matrix inverse:

$$A = U \Sigma V^T$$

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$$A^{-1} = (U \Sigma V^T)^{-1} = (V^T)^{-1} \Sigma^{-1} U^{-1} =$$

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$$A = U \Sigma V^T$$

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$$= V \begin{bmatrix} \frac{1}{\sigma_1} & & \\ & \ddots & \\ & & \frac{1}{\sigma_n} \end{bmatrix} U^T$$

Matrix inverse and solving linear systems

- Matrix inverse:

$$\begin{aligned} A &= U \Sigma V^T \\ A^{-1} &= (U \Sigma V^T)^{-1} = (V^T)^{-1} \Sigma^{-1} U^{-1} = \\ &= V \begin{bmatrix} \frac{1}{\sigma_1} & & \\ & \ddots & \\ & & \frac{1}{\sigma_n} \end{bmatrix} U^T \end{aligned}$$

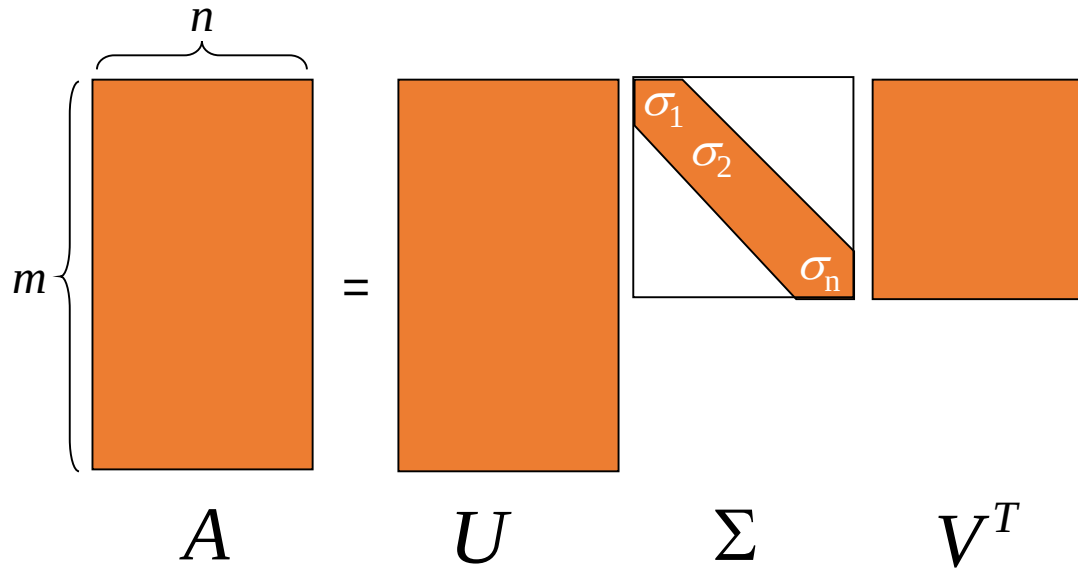
- So, to solve

$$A\mathbf{x} = \mathbf{b}$$

$$\mathbf{x} = V \Sigma^{-1} U^T \mathbf{b}$$

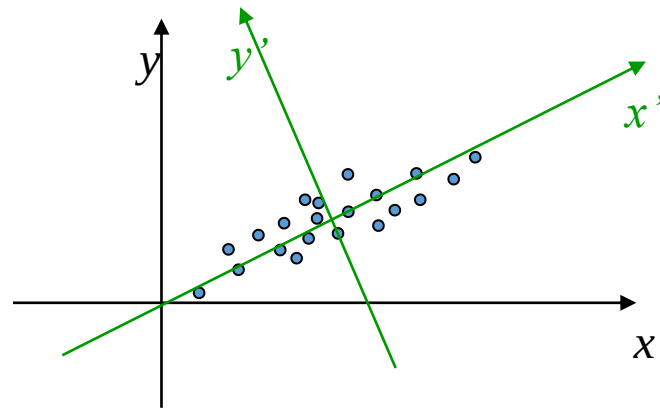
Matrix rank

- The rank of A is the number of non-zero singular values



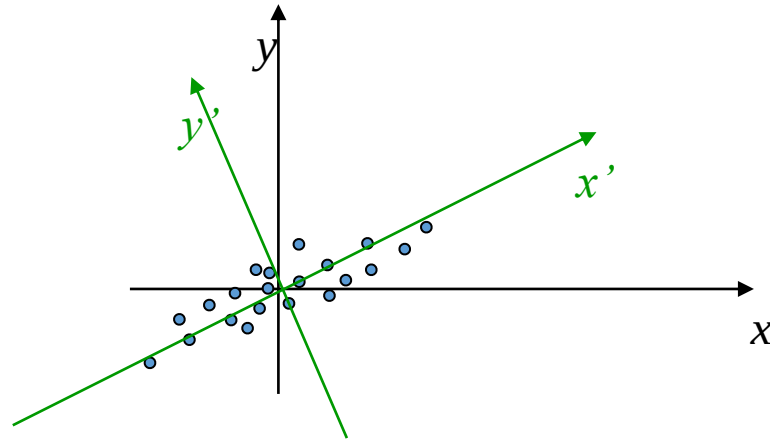
PCA

- We wanted to find principal components



PCA

- Move the center of mass to the origin
 - $\mathbf{p}_i' = \mathbf{p}_i - \mathbf{m}$



PCA

- Constructed the matrix X of the data points.

$$X = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{p}'_1 & \mathbf{p}'_2 & \cdots & \mathbf{p}'_n \\ | & | & \cdots & | \end{bmatrix}$$

- The principal axes are eigenvectors of $S = XX^T$

$$XX^T = S = U \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_d \end{bmatrix} U^T$$

PCA

- We can compute the principal components by SVD of X :

$$X = U\Sigma V^T$$

$$\begin{aligned} XX^T &= U\Sigma V^T (U\Sigma V^T)^T = \\ &= U\Sigma \textcolor{blue}{V}^T \textcolor{blue}{V} \textcolor{red}{\Sigma}^T U^T = U\tilde{\Sigma}^2 U^T \end{aligned}$$

- Thus, the **left singular vectors** of X are the principal components! We sort them by the size of the singular values of X .

Image Compression

- How can we use SVD to compress an image



Image Compression

- How can we use SVD to compress an image

m



n

Image Compression

- How can we use SVD to compress an image



Treat each of the columns of the image as the column of a $m \times n$ matrix

Image Compression

- How can we use SVD to compress an image



$$\begin{pmatrix} \alpha_{1,1} & \alpha_{1,n} \\ \alpha_{m,1} & \alpha_{m,n} \end{pmatrix}_{m \times n}$$

Image Compression

- How can we use SVD to compress an image



m

$$\begin{pmatrix} \alpha_{1,1} & \alpha_{1,n} \\ \alpha_{m,1} & \alpha_{m,n} \end{pmatrix}_{m \times n} = U_{m \times n} \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \end{pmatrix}_{n \times n} V_{n \times n}^T$$

n

Image Compression

- Zero out σ_s up to σ_n



$$\begin{pmatrix} \alpha_{1,1} & \alpha_{1,n} \\ \alpha_{m,1} & \alpha_{m,n} \end{pmatrix}_{m \times n} = U_{m \times n} \begin{pmatrix} \sigma_1 & \cdots & \sigma_n \end{pmatrix}_{n \times n} V_{n \times n}^T$$

S

Image Compression

- Zero out σ_s up to σ_n



$$\begin{pmatrix} \alpha_{1,1} & \alpha_{1,n} \\ \vdots & \vdots \\ \alpha_{m,1} & \alpha_{m,n} \end{pmatrix}_{m \times n} = U_{m \times n} \underbrace{\begin{pmatrix} \sigma_1 & \cdots & \sigma_n \\ \vdots & \times & \vdots \\ \sigma_s & \cdots & \sigma_n \end{pmatrix}}_S_{n \times n} V_{n \times n}^T$$

↓

$$U_{m \times s} \begin{pmatrix} \sigma_1 & \cdots & \sigma_s \\ \vdots & \ddots & \vdots \\ \sigma_s & \cdots & \sigma_s \end{pmatrix}_{s \times s} V_{s \times n}^T$$

Image Compression

- This way your zeroing out the parts that do not have much effect.



$$\begin{pmatrix} \alpha_{1,1} & \alpha_{1,n} \\ \vdots & \vdots \\ \alpha_{m,1} & \alpha_{m,n} \end{pmatrix}_{m \times n} = U_{m \times n} \begin{pmatrix} \sigma_1 & \cdots & \sigma_n \\ \vdots & \times & \vdots \\ \sigma_n & \vdots & \times \end{pmatrix}_{n \times n} V_{n \times n}^T$$

↓

$$U_{m \times s} \begin{pmatrix} \sigma_1 & \cdots & \sigma_s \\ \vdots & \vdots & \vdots \\ \sigma_s & \vdots & \vdots \end{pmatrix}_{s \times s} V_{s \times n}^T$$

Image Compression

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(a)



(b)



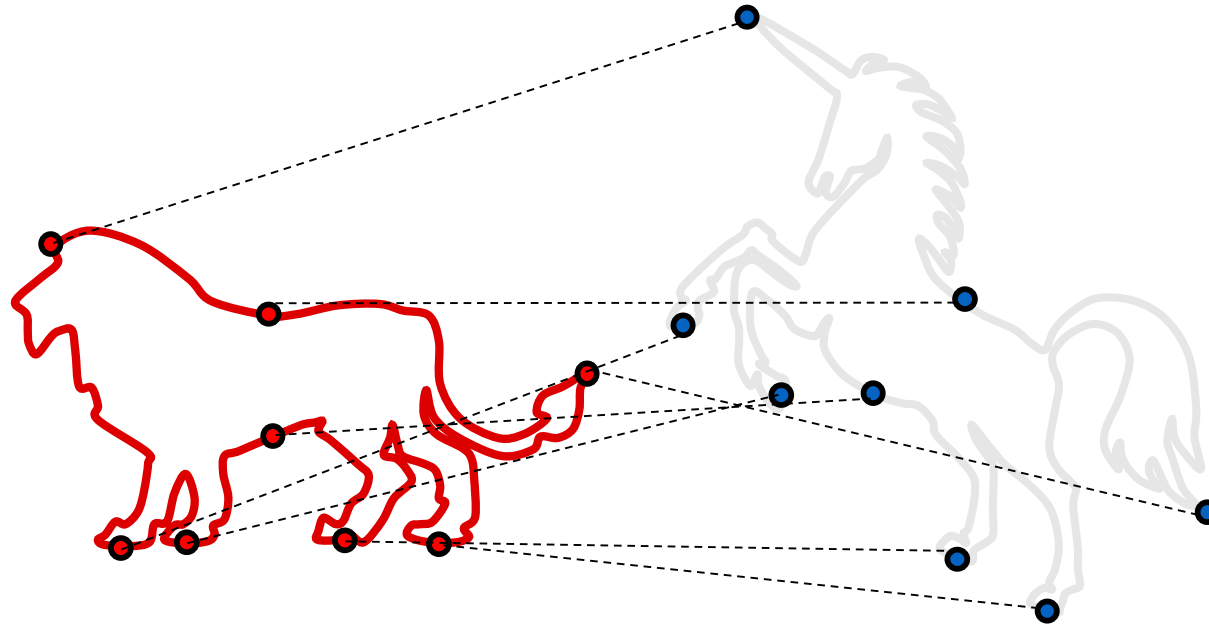
(c)



(d)

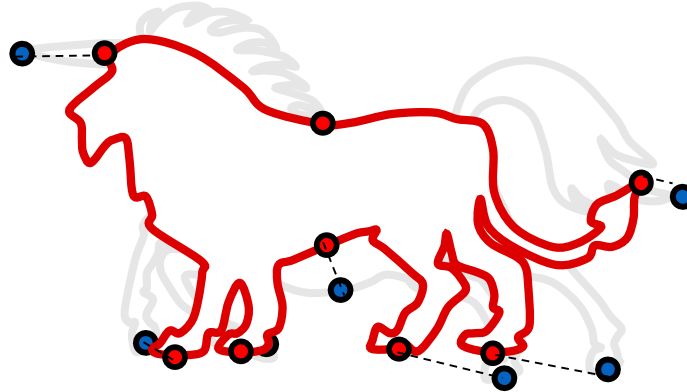
Shape matching

- We have two objects in correspondence
- Want to find the rigid transformation that aligns them



Shape matching

- When the objects are aligned, the lengths of the connecting lines are small.



$P = \{p_1, p_2, \dots, p_n\}$ points in $\mathbb{R}^D$

these two have correspondence

$Q = \{q_1, q_2, \dots, q_n\}$ points in $\mathbb{R}^D$

$P = \{p_1, p_2, \dots, p_n\}$ points in $\mathbb{R}^D$

these two have correspondence

$Q = \{q_1, q_2, \dots, q_n\}$ points in $\mathbb{R}^D$

We are looking for a rigid transformation that maps these two sets of points in a way that the distance between corresponding points would be minimum:

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We are looking for a rigid transformation that maps these two sets of points in a way that the distance between corresponding points would be minimum:

$$(R, t) = \arg \min_{R, t} \sum_{i=1}^n \| (R p_i + t) - q_i \|^2 = F(t)$$

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$$\frac{\partial F}{\partial t} = \sum_{i=1}^n 2 (R P_i + t - q_i)$$

$$= 2 \sum_{i=1}^n R P_i + 2 \sum_{i=1}^n t - 2 \sum_{i=1}^n q_i = 0 \Rightarrow$$

$$t \left(\sum_{i=1}^n 1 \right) = \sum_{i=1}^n q_i - R \sum_{i=1}^n P_i \Rightarrow t = \frac{\sum q_i}{n} - R \frac{\sum P_i}{n}$$

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$$\Rightarrow t = \bar{q} - R \bar{p}$$

The optimal translation t maps the centroid (centre of mass) of transformed p to the centroid of q

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Therefore, define $x_i = p_i - \bar{p}$ and $y_i = q_i - \bar{q}$; move points to the same centroid.

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Therefore, define $x_i = p_i - \bar{p}$ and $y_i = q_i - \bar{q}$; move points to the same centroid.

$$\Rightarrow \text{our expression becomes } R = \underset{R}{\operatorname{argmin}} \sum_{i=1}^n \|Rx_i - y_i\|_2^2$$

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$\Rightarrow$ our expression becomes $R = \underset{R}{\operatorname{argmin}} \sum_{i=1}^n \|Rx_i - y_i\|_2^2$

Let's work with this expression

$$\|Rx_i - y_i\|_2^2$$

$$\|Rx_i - y_i\|_2^2 = (Rx_i - y_i)^T (Rx_i - y_i) = (x_i^T R^T - y_i^T) (Rx_i - y_i)$$

$$\|Rx_i - y_i\|_2^2 = (Rx_i - y_i)^T (Rx_i - y_i) = (x_i^T R^T - y_i^T) (Rx_i - y_i)$$

$$= x_i^T \underbrace{R^T R}_I x_i - \underbrace{x_i^T R^T y_i}_{\text{scalar}} - \underbrace{y_i^T R x_i}_{\text{scalar}} + y_i^T y_i$$

$$\|R\mathbf{x}_i - \mathbf{y}_i\|_2^2 = (R\mathbf{x}_i - \mathbf{y}_i)^T (R\mathbf{x}_i - \mathbf{y}_i) = (\mathbf{x}_i^T R^T - \mathbf{y}_i^T) (R\mathbf{x}_i - \mathbf{y}_i)$$

$$= \mathbf{x}_i^T \underbrace{R^T R}_{\mathbf{I}} \mathbf{x}_i - \underbrace{\mathbf{x}_i^T R^T \mathbf{y}_i}_{\text{scalar}} - \underbrace{\mathbf{y}_i^T R \mathbf{x}_i}_{\text{scalar}} + \mathbf{y}_i^T \mathbf{y}_i$$

$$= \mathbf{x}_i^T \mathbf{x}_i - 2 \mathbf{y}_i^T R \mathbf{x}_i + \mathbf{y}_i^T \mathbf{y}_i$$

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$$= \mathbf{x}_i^T \mathbf{x}_i - 2 \mathbf{y}_i^T R \mathbf{x}_i + \mathbf{y}_i^T \mathbf{y}_i$$

$$\arg \min_R \sum_{i=1}^n \|R\mathbf{x}_i - \mathbf{y}_i\| = \arg \min_R \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{x}_i - 2 \mathbf{y}_i^T R \mathbf{x}_i + \mathbf{y}_i^T \mathbf{y}_i)$$

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$$\arg \min_R \sum_{i=1}^n \|R\mathbf{x}_i - \mathbf{y}_i\| = \arg \min_R \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{x}_i - 2 \mathbf{y}_i^T R \mathbf{x}_i + \mathbf{y}_i^T \mathbf{y}_i)$$

$$\operatorname{argmin}_R \sum_{i=1}^n \|R x_i - y_i\| = \operatorname{argmin}_R \sum_{i=1}^n (x_i^T x_i - 2 y_i^T R x_i + y_i^T y_i)$$

$$= \operatorname{argmin}_R \sum_{i=1}^n (-2 y_i^T R x_i)$$

$$\arg \min_R \sum_{i=1}^n \|R x_i - y_i\|^2 = \arg \min_R \sum_{i=1}^n (x_i^T x_i - 2 y_i^T R x_i + y_i^T y_i)$$

$$= \arg \min_R \sum_{i=1}^n (-2 y_i^T R x_i)$$

Let's consider $n \times n$ $Y^T R X$

$$\arg \min_R \sum_{i=1}^n \|R x_i - y_i\| = \arg \min_R \sum_{i=1}^n (x_i^T x_i - 2 y_i^T R x_i + y_i^T y_i)$$

$$= \arg \min_R \sum_{i=1}^n (-2 y_i^T R x_i)$$

Let's consider $n \times n$ $Y^T R X$

$$Y^T R X = \begin{bmatrix} -y_1^T & - \\ -y_2^T & - \\ \vdots & \\ -y_n^T & - \end{bmatrix}_{n \times d} \begin{bmatrix} R \end{bmatrix}_{d \times d} \begin{bmatrix} | & | & \dots & | \\ x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix}_{d \times n}$$

$$= \underset{R}{\operatorname{argmin}} \sum_{i=1}^n (-2 y_i^T R x_i)$$

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$$= \begin{bmatrix} -y_1^T & - \\ -y_2^T & - \\ \vdots & \\ -y_n^T & - \end{bmatrix} \begin{bmatrix} | & | & \dots & | \\ R x_1 & R x_2 & \dots & R x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} y_1^T R x_1 & y_1^T R x_2 & \dots & \\ y_2^T R x_1 & y_2^T R x_2 & \dots & \\ \vdots & \vdots & \ddots & \vdots \\ y_n^T R x_1 & y_n^T R x_2 & \dots & y_n^T R x_n \end{bmatrix}$$

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$$\sum_{i=1}^n y_i^T R x_i = y_1^T R x_1 + y_2^T R x_2 + \dots + y_n^T R x_n = \underset{\substack{\text{trace} \\ \swarrow}}{\operatorname{tr}} (Y^T R X)$$

$$= \underset{R}{\operatorname{argmin}} \sum_{i=1}^n (-2 y_i^T R x_i)$$

$$Y^T R X = \begin{bmatrix} -y_1^T & - & - \\ -y_2^T & - & - \\ \vdots & & \\ -y_n^T & - & - \end{bmatrix} \begin{bmatrix} | & | & & | \\ R x_1 & R x_2 & \dots & R x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} y_1^T R x_1 & y_1^T R x_2 & \dots & \\ y_2^T R x_1 & y_2^T R x_2 & & \\ \vdots & & \ddots & \\ y_n^T R x_1 & y_n^T R x_2 & \dots & y_n^T R x_n \end{bmatrix}$$

$$\sum_{i=1}^n y_i^T R x_i = y_1^T R x_1 + y_2^T R x_2 + \dots + y_n^T R x_n = \underset{\text{trace}}{\operatorname{tr}} (Y^T R X)$$

$$\Rightarrow \underset{R}{\operatorname{argmin}} \sum_{i=1}^n (-2 y_i^T R x_i) = \underset{R}{\operatorname{argmin}} -2 \operatorname{tr} (Y^T R X) = \underset{R}{\operatorname{argmax}} \underset{\text{trace}}{\operatorname{tr}} (Y^T R X)$$

$$\Rightarrow \arg \min_{\mathbf{R}} \sum_{i=1}^n (-2 \mathbf{y}_i^T \mathbf{R} \mathbf{x}_i) = \arg \min_{\mathbf{R}} -2 \operatorname{tr}(\mathbf{Y}^T \mathbf{R} \mathbf{X}) = \arg \max_{\mathbf{R}} \operatorname{tr}(\mathbf{Y}^T \mathbf{R} \mathbf{X})$$



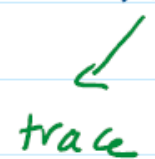
 trace

$$*) \operatorname{tr}(\mathbf{A} \mathbf{B}) = \operatorname{tr}(\mathbf{B} \mathbf{A})$$

$$\Rightarrow \operatorname{tr}(\mathbf{Y}^T \mathbf{R} \mathbf{X}) = \operatorname{tr}(\mathbf{R} \mathbf{X} \mathbf{Y}^T)$$

$d \times d$ Covariance matrix

$$\Rightarrow \arg \min_{\mathbf{R}} \sum_{i=1}^n (-2 y_i^T \mathbf{R} x_i) = \arg \min_{\mathbf{R}} -2 \operatorname{tr}(\mathbf{Y}^T \mathbf{R} \mathbf{X}) = \arg \max_{\mathbf{R}} \operatorname{tr}(\mathbf{Y}^T \mathbf{R} \mathbf{X})$$



trace

$$*) \operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA})$$

$$\Rightarrow \operatorname{tr}(\mathbf{Y}^T \mathbf{R} \mathbf{X}) = \operatorname{tr}(\mathbf{R} \mathbf{X} \mathbf{Y}^T)$$

$d \times d$ Covariance matrix

$$\mathbf{X} \mathbf{Y}^T = \mathbf{S} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T \quad \text{take its SVD}$$

$$\Rightarrow \operatorname{tr}(\mathbf{R} \mathbf{X} \mathbf{Y}^T) = \operatorname{tr}(\mathbf{R} \mathbf{S}) = \operatorname{tr}(\mathbf{R} \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T) = \operatorname{tr}(\mathbf{\Sigma} \mathbf{V}^T \mathbf{R} \mathbf{U})$$

$\textcircled{*}$

we are looking for $\arg\max_Q \text{tr}(\Sigma \underbrace{V^T R U}) = \arg\max_Q \text{tr}(\Sigma M)$

Orthogonal Matrix M

We are looking for $\arg\max_R \text{tr}(\Sigma \underbrace{V^T R U}) = \arg\max_R \text{tr}(\Sigma M)$

Orthogonal Matrix M

Since M is orthogonal for any column m_j , we have $m_j^T m_j = 1$ $m_j = \begin{bmatrix} m_{0j} \\ m_{1j} \\ \vdots \\ m_{dj} \end{bmatrix}$

$$\Rightarrow 1 = m_j^T m_j = \sum_{i=1}^d m_{ij}^2 \Rightarrow m_{ij}^2 \leq 1 \Rightarrow |m_{ij}| \leq 1$$

We are looking for $\arg\max_R \text{tr}(\Sigma \underbrace{V^T R U}_M) = \arg\max_R \text{tr}(\Sigma M)$

Orthogonal Matrix M

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$$\text{tr}(\Sigma M) = \begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_d \end{pmatrix} \begin{pmatrix} m_{11} & m_{12} & \dots & m_{1d} \\ m_{21} & m_{22} & \dots & m_{2d} \\ \vdots & & & \vdots \\ m_{n1} & m_{n2} & \dots & m_{nd} \end{pmatrix} = \sum_{i=1}^d \sigma_i m_{ii} \leq \sum_{i=1}^d \sigma_i$$

$$\text{tr}(\Sigma M) = \begin{pmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \ddots \\ & & & \sigma_d \end{pmatrix} \begin{pmatrix} m_{11} & m_{12} & \dots & m_{1d} \\ m_{21} & m_{22} & \dots & m_{2d} \\ \vdots & & & \vdots \\ m_{n1} & m_{n2} & \dots & m_{nd} \end{pmatrix} = \sum_{i=1}^d \sigma_i m_{ii} \leq \sum_{i=1}^d \sigma_i$$

to get the maximum since M is orthogonal $\Rightarrow M$ must be identity

$$\Rightarrow I = M = V^T R V \Rightarrow V = R V \Rightarrow \boxed{R = V V^T}$$

Summary of rigid alignment:

- Translate the input points to the centroids:

$$\mathbf{p}'_i = \mathbf{p}_i - \mathbf{p}$$

$$\mathbf{q}'_i = \mathbf{q}_i - \mathbf{q}$$

- Compute the “covariance matrix”

$$H = \sum_{i=1}^n \mathbf{q}'_i \mathbf{p}'_i{}^T$$

- Compute the SVD of H :

$$H = U \Sigma V^T$$

- The optimal rotation is

$$\mathbf{R} = V U^T$$

- The translation vector is

$$\mathbf{t} = \mathbf{q} - \mathbf{R} \mathbf{p}$$